EE2005 : TUTORIAL on Induction Machines

1. A 2 - pole, 50 Hz induction motor supplies 15 kW to a load at a speed of 2950 rpm.
   (a) What is the motors slip?
   (b) What is the developed torque in the motor in Nm under these conditions?
   (c) What will the operating speed of the motor be if its torque is doubled?
   (d) How much power will be supplied by the motor when the torque is doubled?

Sol

Given

$P_o = 15\,kW$ (Output) | $f = 50\,Hz$

$N_r = 2950\,rpm$ | $P = 2$

(a) $N_s = \frac{120f}{P} = \frac{120 \times 50}{2} = 3000\,rpm$

$$s = \frac{N_s - N_r}{N_s} = \frac{3000 - 2950}{3000}$$

$$s = 0.0167$$

(b) Mechanical losses are negligable

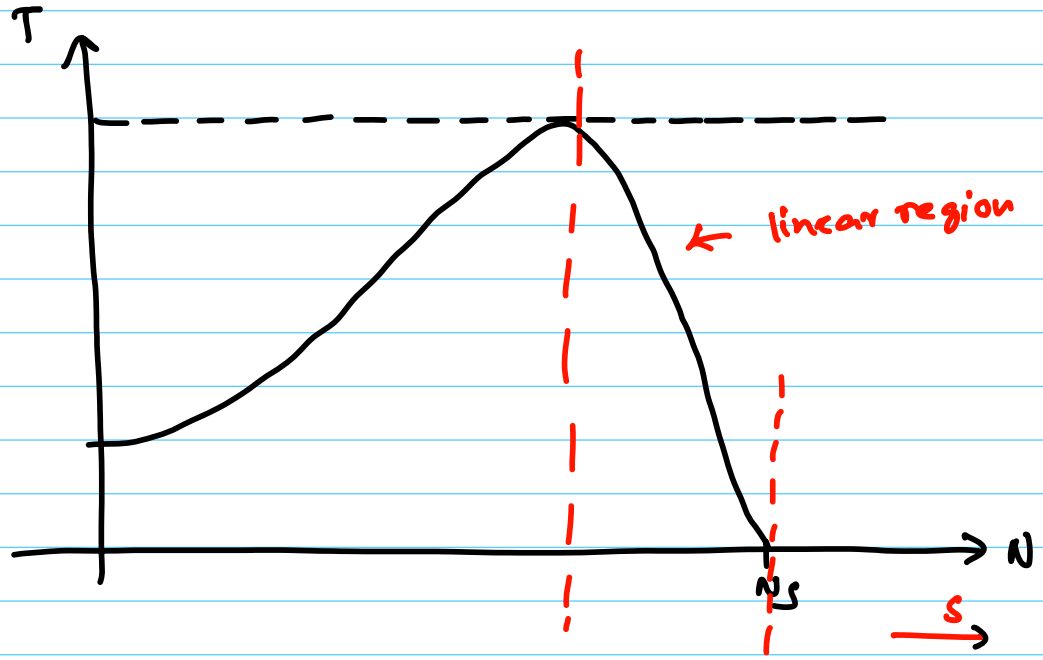
$$P_{conv} \simeq P_o = 15\,kW$$

$$T_{dev} = T_{induced} \simeq \frac{P_o}{\omega_r}$$

$$= \frac{P_o}{\frac{2\pi}{60} \times N_o} = \frac{15 \times 10^3}{\frac{2\pi}{60} \times 2950}$$

$$T_{dev} = 48.5557\,N\text{-}m$$

(c) $T_2 = 2T_1$



$T \propto S$

$$\frac{T_2}{T_1} = \frac{S_2}{S_1}$$

$$\frac{T_2}{T_1} = \frac{\frac{N_S - N_{r2}}{N_S}}{\frac{N_S - N_{r1}}{N_S}}$$

$$\frac{T_2}{T_1} = \frac{N_S - N_{r2}}{N_S - N_{r1}}$$

$$\frac{2T_1}{T_1} = \frac{N_S - N_{r2}}{N_S - N_{r1}}$$

solve for N_{r2}

$$2N_S - 2N_{r1} = N_S - N_{r2}$$

$$N_{r_2} = 2N_{r_1} - N_s$$

$$= 2 \times 2950 - 3000$$

$$N_{r_2} = 2900 \text{ rpm}$$

(d)

$$P_{o_2} = T_2 \times \omega_{r_2}$$

$$= 2 \times T_1 \times \frac{2\pi}{60} \times 2900$$

$$= 2 \times 48.555 \times \frac{2\pi}{60} \times 2900$$

$$P_{o_2} = 29.491 \text{ KW}$$

(or)

$$\frac{P_{o_2}}{P_{o_1}} = \frac{T_2 \times \omega_{r_2}}{T_1 \times \omega_{r_2}}$$

$$= \frac{2T_1 \times N_2}{T_1 \times N_1}$$

$$P_{o_2} = P_{o_1} \times 2 \times \frac{2900}{2950}$$

$$= 2 \times 15 \times 10^3 \times \frac{2900}{2950}$$

$$P_{o_2} = 29.491 \text{ KW}$$

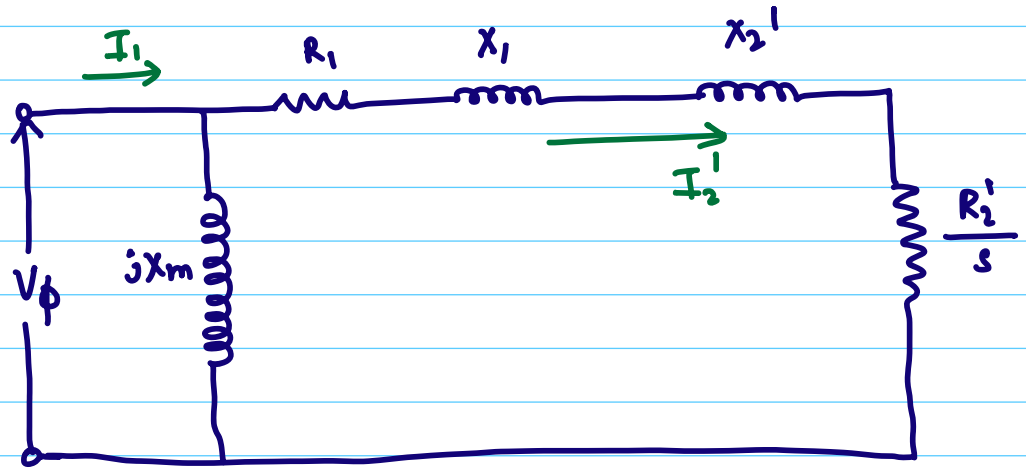
2. A 460 V, 25 hp, 60 Hz, 4-pole, Y-connected induction motor has the following impedances in ohms per phase referred to the stator circuit:
$R_1 = 0.641\,\Omega$, $R_2' = 0.332\,\Omega$, $X_1 = 1.106\,\Omega$, $X_2' = 0.464\,\Omega$, $X_m = 26.3\,\Omega$
The total rotational losses are 1100 W and are assumed to be constant. The core loss is lumped in with the rotational losses. Consider the induction motor in Q.2 ,

   (a) What is the maximum torque of this motor? At what speed and slip does it occur?

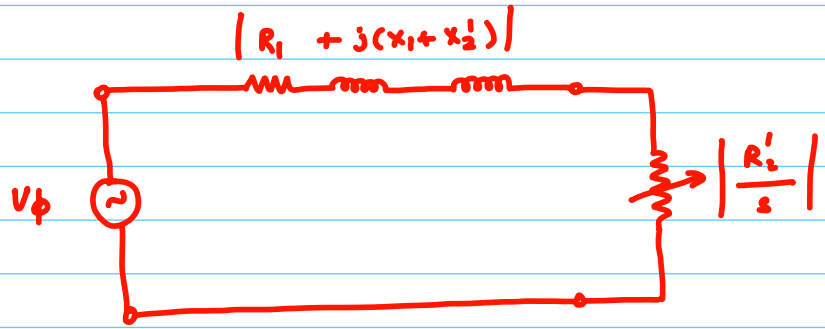
   (b) What is the starting torque of this motor?

   (C) When the rotor resistance is doubled, what is the speed at which the maximum torque now occurs? What is the new starting torque of the motor?

Sol



(a)



Using Maximum Power Transfer Theorem
MPTT

$$\left|\frac{R_2'}{s}\right| = \left|R_1 + j(X_1 + X_2')\right|$$

$s \rightarrow s_{max}$

$$\left| \frac{R_2'}{S_{max}} \right| = \sqrt{R_1^2 + (X_1 + X_2')^2}$$

$$S_{max} = \frac{R_2'}{\sqrt{R_1^2 + (X_1 + X_2')^2}}$$

$$S_{max} = \frac{0.382}{\sqrt{0.641^2 + (1.106 + 0.464)^2}}$$

$$= 0.19577$$

$$S_{max} = 0.19577$$

$$N_r = (1 - S_{max})\, N_s$$

$$N_s = \frac{120\, f}{p}$$

$$= \frac{120 \times 60}{4}$$

$$N_s = 1800 \text{ rpm}$$

$$N_r\,(@\, S_{max}) = (1 - 0.19577)\ 1800$$

$$N_r\,(@\, S_{max}) = 1447.6026 \text{ rpm}$$

$$T_i = \frac{P_{AG}}{\omega_s} \qquad \omega_s = \frac{2\pi N_s}{60} = \frac{2\pi 1800}{60}$$

$$= \frac{3 I_2'^2 \frac{R_2'}{s}}{\omega_s}$$

$$T_i = \frac{3 \times 60}{2\pi N_s} \times I_2'^2 \frac{R_2'}{s}$$

$$T_i = \frac{180}{2\pi N_s} \times I_2'^2 \times \frac{R_2'}{s}$$

@ $s = s_{max}$ $\quad T_i = T_{max}$

$I_2' = ?$

$$|I_2'| = \frac{|V_\phi|}{\sqrt{\left(R_1 + \frac{R_2'}{s}\right)^2 + \left(X_1 + X_2'\right)^2}}$$

$$= \frac{\frac{460}{\sqrt{3}}}{\sqrt{\left(0.641 + \frac{0.332}{s_{max}}\right)^2 + \left(1.106 + 0.464\right)^2}}$$

$$= \frac{\frac{460}{\sqrt{3}}}{\sqrt{\left(0.641 + \frac{0.332}{0.19557}\right)^2 + \left(1.106 + 0.464\right)^2}}$$

$$|I_2'| = 94.336 \text{ A}$$

$$T_{max} = \frac{180}{2\pi \times 1800} \times (94.336)^2 \times \frac{0.332}{0.19557}$$

$$T_{max} = 240.193 \text{ Nm.}$$

(b)

$$s = 1$$

$$I_1' = I_2' \text{ (starting)}$$

$$|I_2'| = \frac{|V_\phi|}{\sqrt{\left(R_1 + R_2'\right)^2 + \left(X_1 + X_2'\right)^2}}$$

$$= \frac{\frac{460}{\sqrt{3}}}{\sqrt{\left(0.641 + 0.332\right)^2 + \left(1.106 + 0.464\right)^2}}$$

$$|I_2'(\text{starting})| = 143.7859 \text{ A}$$

$$T_{st} = \frac{180}{2\pi N_s} \times \left(I_2'(\text{starting})\right)^2 \times \frac{R_2'}{1}$$

$$= \frac{180}{2\pi \times 1800} \times (143.785)^2 \times 0.332$$

$$T_{st} = 109.2422 \text{ Nm}$$

(c)

$$s_{max} = \frac{R_2'}{\sqrt{R_1^2 + (X_1 + X_2')^2}}$$

$$R_2'(ncw) = 2 R_2'$$

$$S_{2,max} = \frac{R_2'(ncw)}{\sqrt{R_1^2 + (X_1 + X_2')^2}}$$

$$= 2 \; s_{max}$$

$$= 2 \times 0.1957$$

$$S_{2,max} = 0.39155$$

$$N_{r2} = (1 - S_{2,max}) \times N_s$$

$$= (1 - 0.39155) \times 1800$$

$$N_{r2} = 1095.205 \text{ rpm}$$

$$T_{st,2} = \frac{180}{2\pi N_s} \times I_2'^2 \times R_2' (\text{new})$$

$$|I_2'| = \frac{|V_\phi|}{\sqrt{(R_1 + 2 \times R_2')^2 + (X_1 + X_2')^2}}$$

$$= \frac{\frac{460}{\sqrt{3}}}{\sqrt{(0.641 + 2 \times 0.332)^2 + (1.106 + 0.464)^2}}$$

$$|I_{2,st}| = 130.088 \text{ A}$$

$$T_{st,2} = \frac{180}{2\pi \times 1800} \times (130.088)^2 \times 2 \times 0.332$$

$$T_{st,2} = 178.839 \text{ Nm}$$

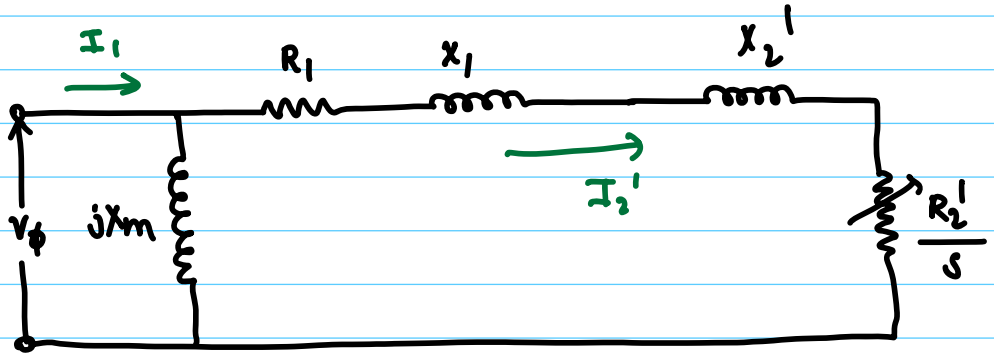
3. A 460 V, 1740 rpm, 60 Hz, 4-pole, wound rotor induction motor has the following parameters per phase:
$R_1 = 0.25\,\Omega$, $R_2' = 0.2\,\Omega$, $X_1 = 0.5\,\Omega$, $X_2' = 0.5\,\Omega$, $X_m = 30\,\Omega$
The rotational losses are 1700 W. With the rotor terminals short-circuited, find

(a) (i) Starting current when started direct on full voltage.
(ii) Starting torque.

((b) (i) Full-load slip.
(ii) Full-load current.
(iii) Ratio of starting current to full-load current.
(iv) Full-load power factor.

Assume approximate equivalent circuit.

Sol



(a)

Starting current = Input current = Current drawn from AC Mains

$$\bar{I}_1 = \frac{V_\phi}{jX_m \,||\, \left((R_1 + R_2'/s) + j(X_1 + X_2') \right)}$$

$$= \frac{V_\phi}{\dfrac{jX_m \left((R_1 + R_2'/s) + j(X_1 + X_2') \right)}{(R_1 + R_2'/s) + j(X_1 + X_2' + X_m)}}$$

$$\bar{I}_1 = \frac{V_\phi \left(\left(R_1 + \frac{R_2'}{s} \right) + j \left(X_1 + X_2' + X_m \right) \right)}{jX_m \left(\left(R_1 + \frac{R_2'}{s} \right) + j \left(X_1 + X_2' \right) \right)}$$

$s = 1$

$$\bar{I}_1 = \frac{\frac{460}{\sqrt{3}} \left((0.25 + 0.2) + j (0.5 + 0.5 + 30) \right)}{j30 \left((0.25 + 0.2) + j (0.5 + 0.5) \right)}$$

$$\bar{I}_1 = 250.288 \angle -66.6039 \text{ A}$$

Input power factor $= \cos \phi$

$= \cos (-66.6039)$

$= 0.397$ lag

$$I_2' = \frac{V_\phi \angle 0}{(R_1 + R_2'/s) + j (X_1 + X_2')}$$

$$= \frac{460/\sqrt{3} \angle 0^\circ}{(0.25 + 0.2) + j (0.5 + 0.5)}$$

$$= 242.189 \angle -65.772^\circ \text{ A}$$

$$|I_2'| = 242.189$$

$$T_{st} = \frac{180}{2\pi \times 1800} \times |I_2'|^2 \times \frac{R_2'}{s}$$

$$= \frac{180}{2\pi \times 1800} \quad 242.189^2 \times 0.2$$

$$T_{st} = 186.7062 \text{ Nm}$$

(b) given rated speed

$$N_r \text{ (full load)} = 1740 \text{ rpm}$$

$$S_{full} = \frac{N_r - N_s}{N_s}$$

$$S_{full} = 0.0333$$

$$\bar{I}_1(full) = \frac{V_p \left(\left(R_1 + \frac{R_2'}{S_{full}}\right) + j\left(X_1 + X_2' + X_m\right)\right)}{jX_m \left(\left(R_1 + \frac{R_2'}{S_{full}}\right) + j\left(X_1 + X_2'\right)\right)}$$

$$= \frac{\frac{460}{\sqrt{3}} \times \left(0.25 + \frac{0.2}{0.033} + j\,81\right)}{j30 \times \left(0.25 + \frac{0.2}{0.033} + j1\right)}$$

$$\bar{I}_1 (full) = 44.230 \angle -20.489^\circ \text{ A}$$

Full load power factor $= \cos(-20.489)$

$= 0.9367$ lag

$$\left| \frac{I_1(st)}{I_1(full)} \right| = \left| \frac{250.288}{44.230} \right|$$

$= 5.6587$

4. A three-phase, 460 V, 60 Hz, six-pole wound-rotor induction motor drives a constant load of 100 Nm at a speed of 1140 rpm when the rotor terminals are short-circuited. It is required to reduce the speed of the motor to 1000 rpm by inserting resistances in the rotor circuit. Determine the value of the resistance if the rotor winding resistance per phase is 0.2 Ω. Neglect rotational losses. The stator-to-rotor turns ratio is unity.

Sol

$$N_s = \frac{120 f}{6}$$

$$= 1200 \text{ rpm}$$

$$\tau \propto \frac{R_2}{s}$$

$$\frac{R_2}{s} = \text{constant}$$

$$R_2'(\text{new}) = R_2' \times \frac{s_1}{s_2}$$

$$= R_2' \times \frac{N_s - N_{r1}}{N_s - N_{r2}}$$

$$= 0.2 \times \left(\frac{1200 - 1000}{1200 - 1140} \right)$$

$$R_2'(\text{new}) = 0.6667$$

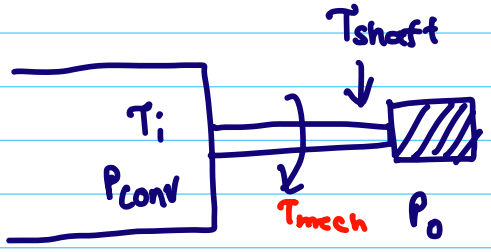
$$R_2' + R_{ext} = 0.6667$$

$$R_{ext} = 0.6667 - 0.2$$

$$R_{ext} = 0.4667 \ \Omega$$

5. A 6-pole, 50 Hz, 3-phase induction motor running on full load develops a useful torque of 162 Nm, and it is observed that the rotor emf makes 90 complete cycles per min. Calculate the power output. If the mechanical torque lost in friction be 13.5 Nm, find the copper loss in rotor windings, the input to the motor and efficiency. The stator losses are 750 W.

Sol



$= 162$ Nm

$T_i = T_{shaft} + T_{mech}$

$= 162 + 13.5$

$= 175.5$ Nm

$T_i = 175.5$ Nm

$f_r =$ 90 cycles per minute

$= \frac{90}{60}$ cycles per sec

$= 1.5$ Hz

$f_r = 1.5$ Hz

$f_s = 50$ Hz

we know, $f_r = sf$

$$s = \frac{f_r}{f}$$

$$= \frac{1.5}{50} = \frac{3}{100}$$

$$s = 0.03$$

$$N_s = \frac{120f}{p}$$

$$= \frac{120 \times 50}{6}$$

$$= 1000 \text{ rpm}$$

$$N_r = (1-s) N_s$$

$$= (1-0.03)\ 1000$$

$$= 970 \text{ rpm}$$

$$P_o = T_{shaft} \times \omega_r$$

$$= 162 \times \frac{2\pi}{60} \times 970$$

$$= 16455.662 \text{ W}$$

$$P_{ag} = T_i \times \omega_s$$

$$= 175.5 \times 1000 \times \frac{2\pi}{60}$$

$$P_{ag} = 18378.317\ W$$

we know

$$P_{ag} : P_{rCL} : P_{conv} = 1 : S : 1-S$$

$$\frac{P_{ag}}{P_{rCL}} = \frac{1}{S}$$

$$P_{rCL} = S\,P_{ag}$$

$$= 0.03 \times 18378.357$$

$$P_{rCL} = 551.349\ W$$

$$P_i = P_{SCL} + P_{ag}$$

$$= 750 + 18378.357$$

$$P_i = 19128.317\ W$$

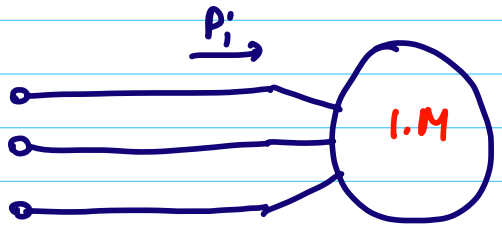
$$\%\,\eta = \frac{P_o}{P_i} \times 100$$

$$= \frac{16455.662}{19128.317W} \times 100$$

$$\%\,\eta = 86.027\ \%$$

6. A 400 V, 50 Hz, 30 hp, 3-ϕ induction motor is drawing 50 A current at 0.8 power factor lagging. The stator and rotor copper losses are 1.5 kW and 900 W respectively. The friction and windage losses are 1050 W and the core losses are 1200 W. Calculate the air-gap power of the motor.

Sol



$$P_i = \sqrt{3}\, V_L\, I_L \cos\phi$$

$$= \sqrt{3} \quad 400 \times 50 \times 0.8$$

$$= \quad 27712.812 \quad W$$

$$P_{SCL} = \quad 1.5\,kW$$

$$P_{RCL} = \quad 900 \quad W$$

$$P_{core} = \quad 1200\ W$$

$$P_{F\&W} = \quad 1050\ W$$

$$P_{ag} = P_{ri} = \quad P_i - P_{SCL} - P_{core}$$

$$= \quad 27712.8129 - 1.5 \times 1000 - 1200$$

$$= \quad 25012.8129\ W$$

7. The blocked rotor test on a 37 kW, 440 V, 50 Hz, 6-pole, Y-connected, 3-ϕ induction motor gave the following results: $V_{line} = 230\,\text{V}$, $I_{line} = 140\,\text{A}$, $\cos\phi = 0.35$. The motor drives a load having a constant torque of 300 Nm. Estimate the maximum percentage reduction in supply voltage possible before the motor stalls. Assume that the copper losses are equally divided between the stator and the rotor. Neglect magnetising current.

Sol

37 KW

440 V Y

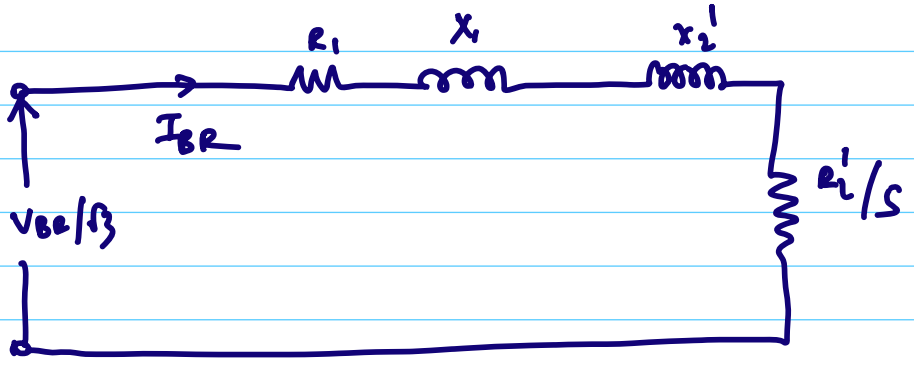
50 Hz

P = 6

Blocked Rotor Test

$V_{BR} = 230\text{V}$ $I_{BR} = 140\,\text{A}$ $\cos\phi = 0.35$

$P_{BR} = \sqrt{3}\, V_{BR} \times I_{BR} \times \cos\phi$



$P_{BR} = \sqrt{3} \times 230 \times 140 \times 0.35$

$= 19520.2126\,\text{W}$

$$P_{SCL} = P_{RCL} = \frac{P_{BR}}{2}$$

$$= 9760.106\,W$$

$$3\, I_{BR}^2 \left(R_1 + R_2'\right) = P_{BR}$$

$$3 \times 2\, I_{BR}^2\, R_1 = P_{BR}$$

$$R_1 = \frac{P_{BR}}{6\, I_{BR}^2}$$

$$R_1 = 0.16598\,\Omega$$

$$R_1 = R_2' = 0.16598\,\Omega$$

$$|Z_{BR}| = \sqrt{(R_1 + R_2')^2 + (X_1 + X_2')^2}$$

$$= \left| \frac{V_{BR}/\sqrt{3}}{I_{BR}} \right|$$

$$= \left| \frac{230/\sqrt{3}}{140} \right|$$

$$= 0.9485\,\Omega$$

$$X_{BR} = \sqrt{Z_{BR}^2 - R_{BR}^2}$$

$$= \sqrt{Z_{BR}^2 - (R_1 + R_2')^2}$$

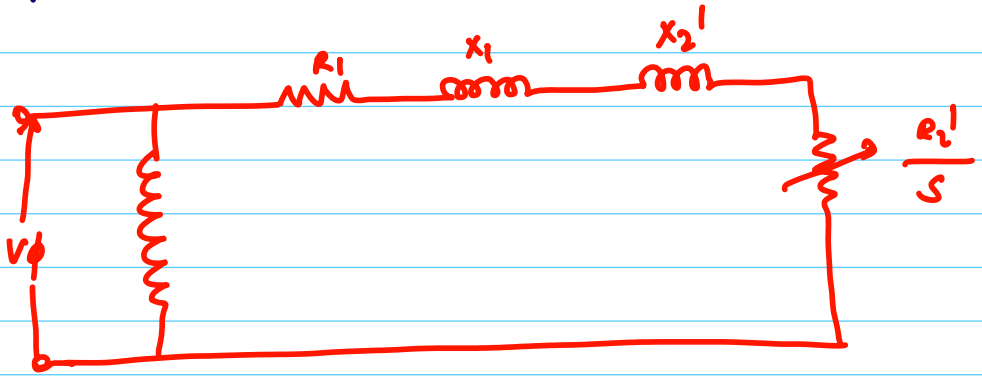
$$= \sqrt{0.9485^2 - 0.33197^2}$$

$$X_{BR} = 0.8885\ \Omega$$

$$X_1 = X_2' = \frac{X_{BR}}{2}$$

$$= \frac{0.8885}{2}$$

$$X_1 = X_2' = 0.4425\ \Omega$$



$$S_{max} = \frac{R_2'}{\sqrt{R_1^2 + (X_1 + X_2')^2}}$$

$$= \frac{0.16598}{\sqrt{0.16598^2 + 0.8885^2}}$$

$$S_{max} = 0.1836$$

$$T_{max} = \frac{180}{2\pi N_S} \times |I_2'|^2 \times \frac{R_2'}{s_{max}}$$

$$|I_2'| = \frac{|V_\phi(ncw)|}{\sqrt{(R_1 + R_2')^2 + (X_1 + X_2)^2}}$$

$$T_{max} = \frac{180}{2\pi N_S} \frac{|V_\phi(ncw)|^2}{(R_1 + \frac{R_2'}{s_{max}})^2 + (X_1 + X_2')^2} \times \frac{R_2'}{s_{max}}$$

Solve for $V_\phi(ncw)$

$$|V_\phi(ncw)|^2 = T_{max} \times \frac{2\pi N_S}{180} \times \frac{s_{max}}{R_2'} \times \left[(R_1 + \frac{R_2'}{s_{max}})^2 + (X_1 + X_2')^2\right]^{\frac{1}{2}}$$

$$|V_\phi(ncw)| = \left[T_{max} \times \frac{2\pi N_S}{180} \times \frac{s_{max}}{R_2'} \times \left[(R_1 + \frac{R_2'}{s_{max}})^2 + (X_1 + X_2')^2\right]\right]^{\frac{1}{2}}$$

$$= \left[300 \times \frac{2\pi \times 1000}{180} \times \frac{0.1836}{0.1659} \left((0.1659 + \frac{0.1659}{0.1836})^2 + 0.8885^2\right)\right]^{\frac{1}{2}}$$

$$= (22407.32)^{\frac{1}{2}}$$

$$|V_\phi(ncw)| = 149.690 \text{ V}$$

$$V_{L-L}(ncw) = \sqrt{3} \; |V_\phi(ncw)|$$

$$= 259.271 \text{ V}$$

$$V_{L-L}(cold) = 440 \text{ V}$$

$$\% \text{ Reduction in Voltage} = \left| 1 - \frac{V_{L-L}(ncw)}{V_{L-L}(cold)} \right| \times 100$$

$$= \left| 1 - \frac{259.271}{440} \right| \times 100$$

$$= 41.074 \%$$

8. An 8-pole, 50 Hz, 3-ϕ induction motor has an equivalent rotor resistance of 0.07 Ω/phase. If its stalling speed is 630 rpm, how much resistance is to be included per phase to obtain the maximum torque at starting? Ignore magnetising current.

Sol

$R_2' = 0.07\ \Omega$

$N_{r,max} = 630$ rpm

$N_s = \frac{120 \times 50}{8}$

$= 750$ rpm

$S_{max} = \frac{750 - 630}{750}$

$= 0.16$

$\frac{R_2'}{S} \propto$ constant

$\frac{R_2'}{S_{max}} = \frac{R_2' \text{ (starting)}}{S \text{ (starting)}}$

$R_2' \text{ (starting)} = R_2' + R_{ext} = R_2' \times \frac{S \text{ (starting)}}{S_{max}}$

$R_{ext} = \left(\frac{S \text{ (starting)}}{S_{max}} - 1 \right) R_2'$

$= \left(\frac{1}{0.16} - 1 \right) \times 0.07$

$R_{ext} = 0.3675\ \Omega$

9. An induction motor has an effiency of 0.9 when the load is 50 hp. At this load, the stator and rotor copper loss each equal the iron loss. The mechanical losses are one-third of the no-load loss. Find the slip.

Sol

$$P_o = 50 \text{ hp}$$

$$= 50 \times 745.7$$

$$P_o = 37.285 \text{ kW}$$

$$\eta = 0.9$$

$$= \frac{P_o}{P_i}$$

$$P_i = \frac{P_o}{0.9}$$

$$= \frac{37.285 \times 10^3}{0.9}$$

$$P_i = 41.427 \text{ kW}$$

$$P_{scl} = P_{RCL} = P_{iron\ (or)\ core}$$

Let no-load loss be

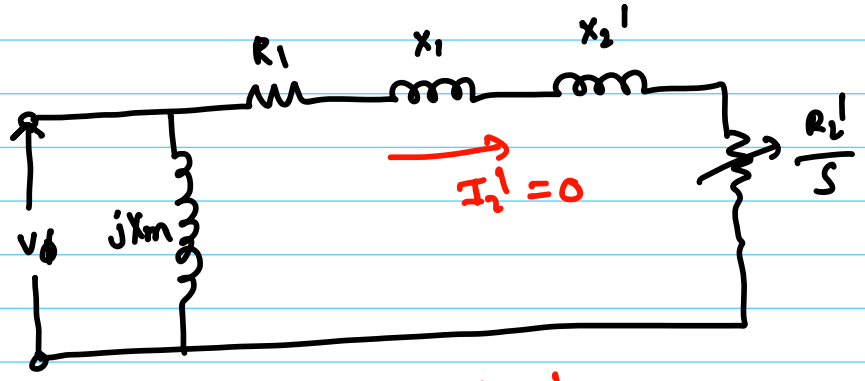
$$P_{no\text{-}load} = y$$

Mechanical losses

$$P_{mech} = \frac{1}{3} \times P_{no\text{-}load}$$

$$= \frac{y}{3}$$

$P_{no\text{-}load}$ = ?



@ no-load

$$\frac{R_2'}{s} \to \infty$$

$$P_{no\text{-}load} = P_{core} + P_{mech}$$

$$y = P_{core} + \frac{y}{3}$$

$$P_{core} = y - \frac{y}{3}$$

$$P_{core} = \frac{2y}{3}$$

$$P_{SCL} = P_{RCL} = P_{core} = \frac{2y}{3}$$

$$P_i = P_o + P_{losses}$$

$$= P_o + P_{SCL} + P_{core} + P_{RCL} + P_{mech}$$

$$= P_o + 3 \times \frac{2y}{3} + \frac{y}{3}$$

$$= P_o + \frac{6y+y}{3}$$

$$P_i = P_o + \frac{7y}{3}$$

$$\frac{7y}{3} = P_i - P_o$$

$$y = \frac{3}{7}\left(\frac{P_i}{P_o} - 1\right) P_o$$

$$= \frac{3}{7}\left(\frac{1}{\eta} - 1\right) P_o$$

$$= \frac{3}{7}\left(\frac{1}{0.9} - 1\right) 50 \times 745.7$$

$$y = 1775.476 \text{ W}$$

So No-load losses $= 1775.476$ W

$$P_{ag} = P_i - P_{SCL} - P_{core}$$

$$= P_i - \frac{2y}{3} - \frac{2y}{3}$$

$$= P_i - \frac{4y}{3}$$

$$P_{ag} = P_i - \frac{4y}{3}$$

$$P_{ag} : P_{RCL} = 1 : S$$

$$\frac{P_{ag}}{P_{RCL}} = \frac{1}{S}$$

$$S = \frac{P_{RCL}}{P_{ag}}$$

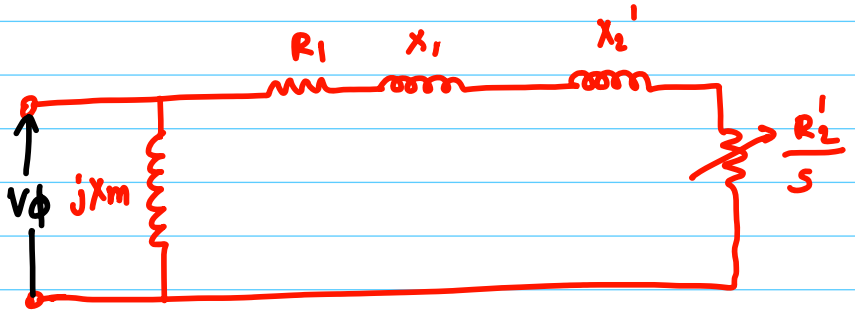
$$= \frac{\frac{2y}{3}}{P_i - \frac{4y}{3}}$$

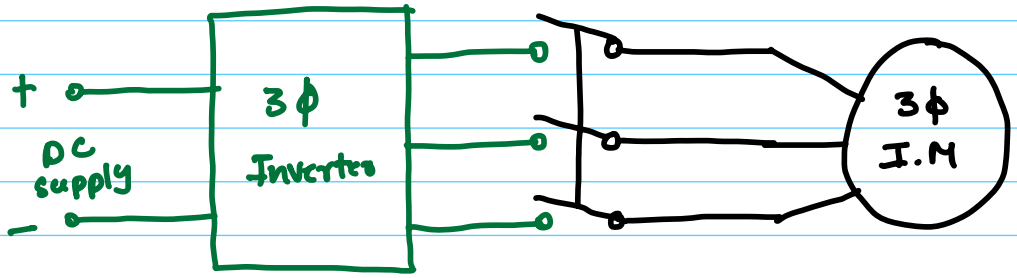
$$= \frac{\frac{2}{3} \times 1775.476}{41.427 \times 10^3 - \frac{4}{3} \times 1775.476}$$

$$S = 0.0303$$

10. A 400 V, 50 Hz, 4-pole, 1400 rpm, Y-connected squirrel cage induction motor has the following parameters referred to the stator: $R_2' = 1.0\,\Omega$, $X_1 = X_2' = 1.5\,\Omega$. Neglect stator resistance and core and rotational losses of the motor. The motor is controlled from a 3-phase voltage source inverter with constant V/f control. Calculate the stator voltage and frequency to obtain the maximum torque at starting. Use the approximate equivalent circuit.

Sol^n





$$\left| \frac{R_2'}{S_{max}} \right| = \sqrt{R_1^2 + (X_1 + X_2')^2}$$

R_1 = negligible

$$\frac{R_2'}{S_{max}} = X_1 + X_2'$$

$$S_{max} = \frac{R_1'}{X_1 + X_1'}$$

$$= \frac{1}{1.5 + 1.5}$$

$$S_{max} = \frac{1}{3}$$

$$f_r = S \times f$$

$$f_r = S_{max} \times f$$

$$f_r = \frac{1}{3} \times 50$$

$$f_r = 16.6667\,Hz$$

$f_s = f_r$ @ max to make torque max. @ starting

But V_ϕ applied is not known.

Given $\frac{V}{f} = \text{constant}$

so $\frac{V_\phi}{f_s} = \frac{V_\phi\,(new)}{f_r\,(new)}$

$$V_{\phi(new)} = V_{\phi} \frac{f_{s(new)}}{f_s}$$

$$= \frac{400}{\sqrt{3}} \times \frac{f_s/3}{f_s}$$

$$V_{\phi(new)} = 76.98\ V$$

$$V_{L-L(new)} = V_{\phi(new)} \sqrt{3}$$

$$= 133.333\ V$$